

Natanael Detamor Karo Karo

Full-Stack Developer

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SUMMARY

Full-Stack Developer with 1 year of production experience building ERP, HRIS, and AI-driven systems using Vue.js, Laravel, and MySQL, now extending the stack with React.js, Go, and the MERN stack. Designed and shipped a production ERP actively used by 32 employees, independently built a 6-service e-commerce backend in Go, and developed a full-stack MERN donation platform (React, Node.js/Express, MongoDB) for a children's foundation, including dual payment gateway integration (Midtrans, Xendit) and a hardened security pipeline. Comfortable across the full stack, from React/Vue interfaces to Go/Node/PHP APIs and relational/NoSQL database design.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Go (Golang), PHP, Python, SQL

Frontend: React.js, Vue.js, Tailwind CSS, Bootstrap, Framer Motion, Vite, jQuery

Backend: Go (Gin Gonic, GORM), Node.js (Express.js), Laravel, FastAPI, REST API Design

Databases & Caching: PostgreSQL, MongoDB (Aggregation), MySQL, Redis

DevOps & Tools: Docker, Docker Compose, Nginx, Git, GitHub Actions (CI/CD), RabbitMQ, Postman

Practices: Microservices Architecture, JWT Authentication, Security Middleware, Unit Testing, Agile Workflow

EXPERIENCE

Full-Stack Developer — Intern & Freelance | Hope Channel Indonesia Jun 2025 – May 2026

Internship: Jun–Aug 2025 | Freelance: Sep 2025–May 2026 | Laravel, Vue.js, MySQL, Tailwind CSS, jQuery

- Designed full system architecture of Hopemedia.id ERP from scratch (Vue.js dashboards + Laravel REST API + MySQL); collaborated directly with the Program Manager to align the system with operations across 3 teams (Program, Broadcasting, HR).
- Built a fingerprint attendance system that auto-parses TXT exports from biometric hardware, plus Zoom API integration for activity tracking; the system serves 32 employees with role-specific dashboards across 22 roles.
- Architected an 18-phase broadcast production workflow (freelance) with automated deadline tracking and KPI scoring, plus an HRIS with 3 independent approval hierarchies and dynamic leave quotas.
- Implemented JWT authentication with auto-logout and session management, CSP security headers, and Google Calendar API integration; improved performance via lazy loading, API caching, and N+1 query resolution; covered business logic with PHPUnit tests.

PROJECTS

Scalable E-Commerce Platform — Go Microservices Backend | Personal Project 2026

Go, Gin Gonic, GORM, PostgreSQL, Redis, RabbitMQ, Docker, Nginx | github.com/detamor/Scalable-E-Commerce-Platform

- Built a 6-service microservices backend in Go (user, product, cart, order, payment, notification) with clean/layered architecture, containerized with Docker Compose behind an Nginx API gateway.
- Implemented a stateful Circuit Breaker and exponential-backoff retry from scratch in Go (thread-safe via sync.Mutex), plus a custom idempotent migration system; tested with mock repositories and a GitHub Actions CI/CD pipeline.

YAPI Medan — Donation & Social Platform (Full-Stack MERN)

Aug 2025 – Present

Volunteer Project for a children's foundation | React 18, TypeScript, Node.js, Express.js, MongoDB | github.com/detamor/YapiFe + [Yapi_medanBE](#)

- Built a full-stack MERN platform: a React + TypeScript frontend consuming a Node.js/Express/MongoDB REST API across modules for auth, donations, children profiles, activities, and admin reporting.
- Integrated dual payment gateways (Midtrans Snap, Xendit Invoices) with webhook signature verification for automated donation status updates (currently in sandbox/test mode), plus MongoDB aggregation pipelines for real-time reporting.
- Implemented JWT authentication with RBAC and an admin-approval flow; hardened the API (Helmet, mongo-sanitize, xss-clean, hpp, rate limiting) and encrypted client-side tokens with AES (CryptoJS) plus DOMPurify sanitization.

Expert System — Plant Disease Diagnosis

Nov 2025 – Apr 2026

Final Year Research Project · Team of 3 · Role: Lead Architect & AI Engine | Laravel, FastAPI, Python, Vue.js, MySQL

- Designed a 3-service microservice architecture and built the diagnostic engine using Forward Chaining + Certainty Factor (50% threshold), with an admin knowledge base, OTP authentication, and downloadable PDF diagnosis reports.

EDUCATION

Bachelor of Informatics Engineering — Universitas Advent Indonesia

Aug 2022 – Jun 2026

GPA: 3.61/4.00 | OOP, Data Structures, DBMS, Web Dev I & II, Cloud Computing, Artificial Intelligence, Business Intelligence

CERTIFICATIONS

- IT Specialist: Databases — Certiport (Mar 2025)
- Structured Query Language (SQL) — Dicoding (Apr 2025)
- Javascript Programming – Dicoding (Jan 2025)
- Backend – With Javascript – Dicoding (Jan 2025)